TeAPOT: Smartphone-Assisted Terrain-Aligned Geospatial Registration for Head-Mounted Mixed Reality in Outdoor Environments

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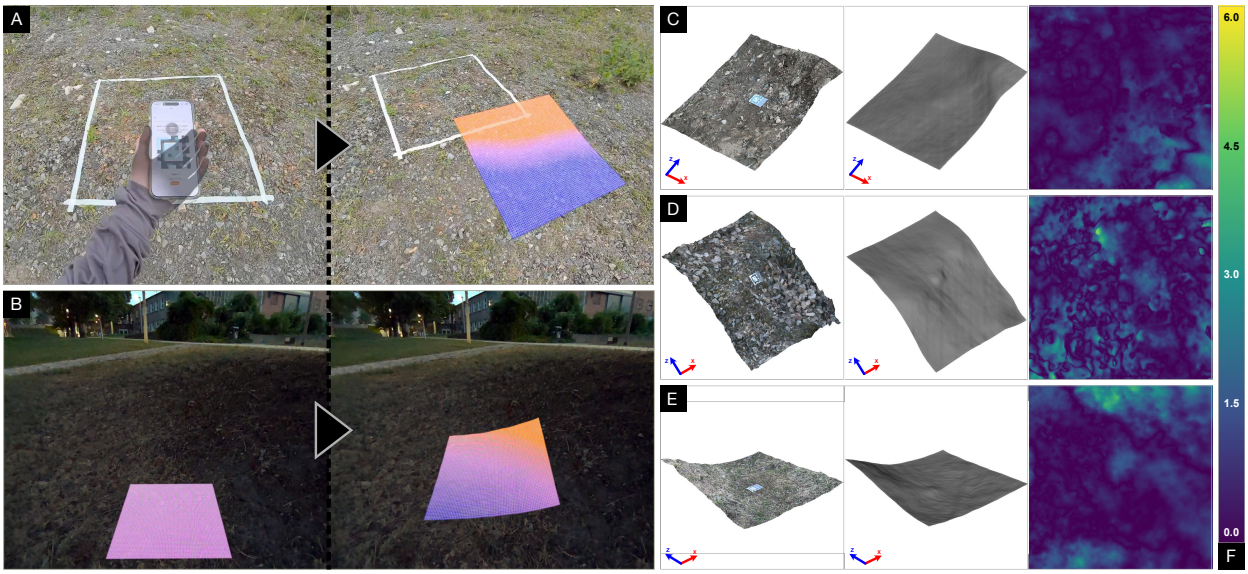


Figure 1: (A) TeAPOT in use: smartphone-assisted geospatial registration and the resulting terrain-conforming surface, with visible placement error relative to the tape-marked ground-truth location. (B) Local terrain reconstruction at another location: the flat proxy plane before deformation, shown only for diagnostic comparison, and the resulting terrain-conforming surface. (C–E) Rows show bare soil, mixed terrain, and grass. Columns show the real-terrain mesh, the reconstructed terrain rendered with an Eye Dome Lighting (EDL) shader in CloudCompare, and the absolute height-difference map. (F) Height-difference color scale (0.0–6.0 cm) used in C–E.

Abstract— Some outdoor mixed reality (MR) applications, such as exploratory visualization of smoothly varying environmental fields, tolerate meter-scale global placement error but still require spatially extended overlays to conform to the local terrain. However, standalone MR headsets often lack Global Navigation Satellite System (GNSS) receivers and magnetometers, and flat-ground assumptions can cause overlays to float above or intersect uneven terrain. We present TeAPOT (Terrain-Aligned Positioning and Orientation Toolkit), which transfers smartphone GNSS position and true heading to a headset through fiducial alignment, then reconstructs a local 2.5D terrain patch on the headset from stereo environment-depth observations for terrain-aligned rendering. Across 150 outdoor trials at three locations and standing distances up to 10 m, TeAPOT achieved a median end-to-end horizontal placement error of 2.04 m ($P_{95} = 5.52$ m), with end-to-end error strongly correlated with the corresponding standing-position error. A separate 60-session characterization yielded a median absolute true-heading error of 4.08° ($P_{95} = 8.19°$), consistent with the observed increase in added error at larger standing distances. For local terrain reconstruction, 30 reconstructions of 1×1 m patches across three terrain types yielded a median absolute height error of 0.6 cm ($P_{95} = 1.1$ cm) relative to reference scans of the same patches. The median time to stability was 1.1 s, and the median GPU time was 5.6 ms. These results demonstrate meter-tolerant outdoor MR on commodity hardware, combining meter-scale absolute placement with centimeter-level local surface-shape accuracy.

Index Terms—Geospatial mixed reality, outdoor mixed reality, spatial alignment, positioning and orientation, terrain reconstruction.

1 Introduction

Outdoor mixed reality (MR) supports field applications in which users explore and analyze spatial content on-site. Examples include immersive drone operation [29], cyber-physical games [2], and situated visualization of environmental fields and subsurface infrastructure [5, 10]. Registration requirements vary by application: infrastructure inspection may require survey-grade accuracy, whereas exploratory visualization of area-scale environmental fields can tolerate meter-scale global error. Variables such as groundwater-table depth, soil moisture, and vegetation health are often represented as continuous fields interpolated from sparse measurements [34, 35, 37]. When these fields vary smoothly over distances comparable to or greater than the global placement error, querying the field at the estimated rather than the true location may

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preserve the same qualitative interpretation [18, 28].

Placing virtual content at geographic locations requires geospatial registration: the headset must know its position and orientation in a global coordinate frame. Smartphone augmented reality (AR) SDKs already support geographic anchoring, for example through ARKit Location Anchors[1] and the ARCore Geospatial API[2]. In supported areas, such services can combine device sensing with map-based visual localization, as in Google's Visual Positioning System (VPS) in the ARCore Geospatial API. However, such systems depend on prebuilt and sufficiently current visual maps, which may be unavailable in natural, remote, or rapidly changing field sites.

In supported environments, smartphone-based geospatial anchoring makes handheld AR viable. By contrast, head-mounted MR is useful when users navigate uneven terrain while inspecting or configuring situated visualizations. It provides hands-free operation, binocular depth cues, and gesture-based interaction for tasks such as switching data layers or selecting time steps. Standalone headsets, however, often lack integrated Global Navigation Satellite System (GNSS) receivers and magnetometers. Consequently, outdoor head-mounted systems commonly rely on specialized positioning hardware, additional sensors, external tracking, or installed infrastructure, limiting deployability.

Geospatial registration alone is insufficient for spatially extended, ground-attached content, because the system must also represent the local terrain. Flat-ground assumptions can cause overlays to float above or intersect uneven surfaces, while existing elevation or surface models may be unavailable, outdated, or too coarse for local alignment. Terrain alignment is especially important for situated visualization of spatial data tied to the ground surface, because the terrain provides the spatial frame through which values are perceived and understood [7, 27].

Current consumer hardware makes a lightweight solution practical. Some modern smartphones provide dual-frequency GNSS position and true-heading estimates, while some extended reality (XR) headsets expose camera feeds and environment-depth estimates for local terrain reconstruction. When only a local ground patch is required, a 2.5D elevation representation avoids the cost of full volumetric or dense-mesh reconstruction while supporting terrain-relative placement and ground-aligned overlays. We therefore present TeAPOT (Terrain-Aligned Positioning and Orientation Toolkit), which combines smartphone-assisted geospatial registration with on-headset terrain reconstruction for outdoor MR applications that tolerate meter-scale global error and do not require survey-grade registration.

Our key contributions are:

- **Smartphone-assisted geospatial registration**: A phone-headset registration pipeline that transfers smartphone GNSS position and true heading through fiducial-based alignment and headset tracking.
- **On-headset terrain reconstruction**: A lightweight depth-to-height-field pipeline for terrain-aligned placement of spatially extended virtual content.
- **Prototype and evaluation**: A prototype implementated using off-the-shelf hardware and an outdoor evaluation covering smartphone position and true-heading errors, end-to-end registration, local terrain accuracy, and runtime performance.

Evaluation data, analysis scripts, supplementary videos, and the authors' version of this paper with all appendices are available at doi.org/10.5281/zenodo.21468153. TeAPOT source code is publicly available at github.com/PiBu-Labs/TeAPOT.

2 Related Work

2.1 Geospatial Registration in Mixed Reality

Situated visualization in AR places data in a direct spatial relationship with physical referents [13]. For geospatially referenced outdoor content, this requires estimating a global pose. Early georegistered outdoor AR systems supported urban exploration, site visits, and infrastructure visualization, including the head-mounted Touring Machine [9] and handheld systems such as SiteLens [32] and SmartVidente [23]. Smartphone-based outdoor AR systems have also combined object detection with Global Positioning System (GPS), Inertial Measurement Unit, and magnetometer for registration and mobile geovisualization [20]. However, these systems either predate modern standalone headsets or focus on handheld use.

For outdoor head-mounted AR and MR, high-accuracy global registration has commonly relied on specialized sensing and multi-sensor fusion. Prior systems use differential GNSS, including Real-Time Kinematic (RTK), often augmented with magnetometers, inertial sensors, and barometric altimeters [3, 10, 14, 22, 26]. These systems can achieve higher accuracy than phone-only commodity sensing but typically require additional hardware, calibration, correction services, or infrastructure.

Another approach is map-based visual localization, in which camera observations are matched against prebuilt visual maps. Stehlin et al. use a smartphone running the ARCore Geospatial API to transfer a global pose to a Microsoft HoloLens 2 [25]. Such approaches can be advantageous in mapped urban environments, where buildings may degrade GNSS, but depend on visual-map coverage, recognizable features, and sufficiently current maps. TeAPOT instead targets natural or changing outdoor environments where such maps may be unavailable or unreliable.

A commodity alternative uses the phone's own positioning and heading sensors. Studies of dual-frequency GNSS in consumer smartphones report improved positioning, although performance remains dependent on the environment, antenna quality, and observation time [21, 30, 33, 36]. Smartphone compass measurements can likewise exhibit direction- and environment-dependent offsets that directly affect visual registration [4]. TeAPOT uses a dual-frequency smartphone as its sole external sensing device and transfers its GNSS position and reported true heading through fiducial-based phone-headset alignment. It therefore targets meter-tolerant outdoor MR without RTK hardware or prebuilt visual maps.

2.2 Terrain-Aligned Surface Reconstruction

Global placement alone is insufficient for outdoor situated visualizations whose content must conform to the local ground surface. Reviews of AR for subsurface utilities identify positional error, depth perception, and inadequate surface handling as challenges [17]. Prior surface-aware systems use pre-captured road meshes to drape utility information [6] or ground geometry to align and occlude utility excavation models [10, 11]. These approaches support surface-aware visualization but rely on geometry captured or reconstructed in advance.

More generally, XR scene meshes and live depth maps support occlusion and content placement but typically provide general-purpose scene representations rather than a dedicated reconstruction of a small local ground patch. Dense fusion methods such as KinectFusion [19] reconstruct full 3D geometry and are therefore more general than required here. When the target surface can be represented by a single height per ground-plane cell, a height field provides a lighter representation. Watson et al. discuss height fields as efficient reconstruction targets for AR [31]. Similar representations are used in robotics, where local elevation maps support terrain representation and terrain-aware reasoning [1, 8]. TeAPOT adopts this representation to reconstruct a local 2.5D terrain patch for terrain-aligned rendering rather than navigation.

TeAPOT's fusion strategy draws on point-based reconstruction from irregular samples. Surface splatting and Elliptical Weighted Average (EWA) filtering use elliptical sample footprints to reduce aliasing and improve surface continuity [38]. We adapt this principle by projecting headset depth samples onto a proxy ground plane and fusing their height contributions over time.

2.3 Multi-Device Spatial Alignment

TeAPOT also relates to spatial alignment across devices. Co-located multi-user MR, for example, requires transformations between disparate device coordinate spaces [15]. ViewR addresses large-scale alignment and recovery for co-located mobile MR headsets in architectural-scale experiences [24]. Recent work has also used fiducial tags as a

[1] https://developer.apple.com/documentation/arkit

[2] https://developers.google.com/ar/develop/geospatial

local alternative to cloud-based shared anchors for aligning multiple XR devices in a common space [16]. These approaches establish shared local frames rather than transferring an absolute geospatial frame to a headset.

The work by Stehlin et al. [25] is the most directly related because it also uses a smartphone to transfer a global pose to a headset. TeAPOT differs by using phone GNSS and reported true heading rather than a VPS-derived pose, transferring them through fiducial alignment, and adding on-headset terrain reconstruction.

3 Methodology

In this section, we describe TeAPOT's processing pipeline. We use a left-handed coordinate system with $+X$ pointing right, $+Y$ up, and $+Z$ forward. Throughout, ${}^{A}\mathbf{T}_{B}$ denotes the rigid transform that maps coordinates from frame B to frame A.

3.1 Geospatial Registration

To anchor virtual content at geographic locations, we first determine the geospatial pose of the virtual-world origin: its geodetic position (latitude and longitude) and true heading. The origin's geodetic position is derived from the phone's GNSS, while its heading is obtained by combining the phone's compass heading with the headset's observation of a fiducial tag displayed on the phone, which determines the phone pose in the virtual-world frame.

Phone pose in the virtual world Let P denote the phone reference frame and T the displayed tag frame. The phone pose in the virtual world is obtained by chaining four available transforms:

$$ {}^{V}\mathbf{T}_{P} = {}^{V}\mathbf{T}_{H}\,{}^{H}\mathbf{T}_{C}\,{}^{C}\mathbf{T}_{T}\left({}^{P}\mathbf{T}_{T}\right)^{-1}, \tag{1} $$

where ${}^{V}\mathbf{T}_{P}$ denotes the phone pose in the virtual world, ${}^{V}\mathbf{T}_{H}$ the headset pose in the virtual world, ${}^{H}\mathbf{T}_{C}$ the camera pose in the headset frame, ${}^{C}\mathbf{T}_{T}$ the tag pose in the camera frame, and ${}^{P}\mathbf{T}_{T}$ the tag pose in the phone frame.

Geodetic position of the virtual-world origin The phone's geodetic position is determined from its GNSS readings, expressed as latitude and longitude. The translation component of ${}^{V}\mathbf{T}_{P}$ from Eq. (1) gives the offset from the virtual-world origin to the phone. Using the origin heading derived below, we transform the horizontal component of this offset from the virtual frame to the local east–north coordinates. We then recover the virtual-world origin by subtracting the transformed offset from the phone's geodetic position. In our implementation, these World Geodetic System 1984 (WGS84) coordinate conversions use Earth-Centered, Earth-Fixed (ECEF) as an intermediate representation. For these conversions, we assign both the phone and the virtual-world origin an altitude of zero, i.e., we project them onto the WGS84 reference ellipsoid. This is acceptable for our use case because we only use the resulting coordinates to compute relative placement over short baselines, which are at most about 10 m for the proxy-plane placement. Over such distances, the horizontal error introduced by ignoring the true altitude is negligible compared with the measured meter-scale GNSS error and the potential distance-dependent effect of heading error.

True heading of the virtual-world origin The phone's compass provides a true heading ψ_P, defined as the clockwise angle from true north to the phone's top-edge direction in portrait coordinates. The top-edge direction in the virtual world is

$$ \mathbf{u} = {}^{V}\mathbf{R}_{P}\,[0,\,1,\,0]^{\top} = [u_x,\,u_y,\,u_z]^{\top}, $$

where ${}^{V}\mathbf{R}_{P}$ is obtained from Eq. (1). Its yaw relative to the virtual-world $+Z$ axis is

$$ \theta_{\text{yaw}} = \operatorname{atan2}(u_x,\,u_z) \bmod 2\pi, $$

i.e., the clockwise angle from the virtual-world $+Z$ axis to the phone's top-edge direction. Assuming the nearby phone and virtual-world origin share the same local true north, the origin heading is

$$ \psi_O = (\psi_P - \theta_{\text{yaw}}) \bmod 2\pi. $$

Phone-headset geospatial registration Each tag detection is paired with the temporally nearest phone observation. The resulting origin-heading estimates are first aggregated using a circular mean. The mean heading is then used to recover an origin-position estimate from each paired observation, and these position estimates are aggregated using a spherical mean.

3.2 Proxy Plane Placement

Once the geospatial pose of the virtual-world origin is known, we convert the target latitude and longitude to ECEF coordinates using the same zero-altitude assumption. We subtract the origin position and transform the resulting offset into the virtual frame. This gives the target position relative to the virtual-world origin, at which we place the planar proxy with its normal pointing upward. Local terrain reconstruction then deforms the proxy to match the observed ground surface at that location.

3.3 Local Terrain Reconstruction

Because the origin and target are assigned the same ellipsoidal height, geospatial placement omits their local terrain-height difference. The proxy may therefore appear to float above or intersect the ground. To address this, we use the headset's per-eye environment-depth maps to estimate the local ground surface.

We generate a 2.5D height field by forward-splatting unprojected depth samples onto a local ground-aligned grid on the proxy plane. Because these samples are noisy and irregularly distributed on the plane, we use an EWA-style elliptical kernel [38], so that each sample contributes to multiple nearby grid cells rather than only its nearest cell. Since the target is a ground-aligned height field, the splat footprint is expressed in proxy-plane coordinates rather than derived from estimated per-sample normals. Splatting accumulates a weight and weighted height in each grid cell. Samples with excessively grazing incidence angles are rejected to suppress unreliable contributions.

We then apply temporal fusion by updating each cell's persistent height estimate with a capped running weighted average, following the weighted running average update used in KinectFusion [19]. This prevents old observations from dominating indefinitely while allowing the estimate to adapt to later measurements.

For each depth pixel at image coordinates $(u,\,v)$ with depth z, we unproject it to 3D world space and project it onto the local ground plane. Let $\mathbf{p}$ denote the unprojected world point, and let $\mathbf{o}$, $\mathbf{r}$, $\mathbf{f}$, and $\mathbf{n}$ denote the proxy-plane origin, right, forward, and normal vectors, respectively. The sample's in-plane coordinates and height are

$$ \mathbf{c} = \begin{bmatrix} (\mathbf{p}-\mathbf{o})\cdot\mathbf{r} \\ (\mathbf{p}-\mathbf{o})\cdot\mathbf{f} \end{bmatrix}, \qquad h_s = (\mathbf{p}-\mathbf{o})\cdot\mathbf{n}. $$

We estimate the sample footprint on the plane from pixel differentials using the Jacobian $\mathbf{J} = [\mathbf{a},\,\mathbf{b}]$, whose columns are the projections of $\partial\mathbf{p}/\partial u$ and $\partial\mathbf{p}/\partial v$ onto the plane axes. We model the footprint as an ellipse with precision matrix

$$ \mathbf{C} = \tau^{-2}\left(\mathbf{J}\mathbf{J}^{\top}\right)^{-1}, $$

where τ controls the Gaussian spread. To avoid holes in undersampled regions, we clamp each column of $\mathbf{J}$ to a minimum length of half the grid-cell size. Let $A_{\text{cell}} = \Delta c_1 \Delta c_2$ denote the area of a grid cell. For a grid cell centered at offset $\mathbf{d}_{ij}$ from the sample center, the discrete EWA weight is

$$ w_{ij} = A_{\text{cell}}\,\frac{1}{2\pi\tau^2|\det(\mathbf{J})|}\,\exp\!\left(-\frac{1}{2}\mathbf{d}_{ij}^{\top}\mathbf{C}\mathbf{d}_{ij}\right), $$

evaluated within a finite elliptical support. Each accepted sample contributes to

$$ S = \sum w_{ij}, \qquad H = \sum w_{ij}\,h_s. $$

Temporal fusion is then applied per cell over time. From the current terrain update's spatial accumulators, we compute

$$ h_{\text{frm}} = \frac{H}{S}, \qquad w_{\text{frm}} = \operatorname{saturate}(S), $$

where h_{frm} is the current-update height estimate and w_{frm} is its support-derived weight. Each cell stores a fused height mean μ and an effective weight W, updated as

$$W' = \min(W + w_{\text{frm}}, W_{\max}), \qquad \mu' = \mu + \frac{w_{\text{frm}}}{W + w_{\text{frm}}}(h_{\text{frm}} - \mu),$$

with $W_{\max} = 30$. While $W < W_{\max}$, this behaves as an online weighted running mean. Once W reaches $W_{\max}$, it becomes a bounded-step exponential smoother. Cells are initialized on first observation, and when no measurement arrives, the previous estimate is retained without decay. The fused height field is then used to deform the proxy plane into terrain-aligned geometry.

4 Implementation

4.1 System Overview

TeAPOT comprises a mobile app and a head-mounted MR app. The phone provides GNSS position and true heading and displays a fiducial tag for phone-headset alignment. The headset estimates the phone pose from the tag, establishes the geospatial frame, reconstructs the local terrain, and renders terrain-aligned content. The phone is required only during initial registration.

The mobile app runs on an iPhone 16 Pro Max and is implemented in Swift with SwiftUI. It obtains GNSS position, true heading, and their uncertainty estimates from Core Location. Once `horizontalAccuracy` remains at or below 2.5 m for 5 consecutive fixes, the app displays an AprilTag from the `tagStandard41h12` family [12] and begins streaming timestamped position and heading via WebSocket. The tag size, defined as the distance between adjacent detection corners, is 3 cm. The app uses a device-specific pixels per inch lookup and the display scale to render this physical distance. On the test device, it corresponds to approximately 543 native display pixels, which we verified with a ruler. During handheld registration, the tag was observed from approximately 30–60 cm.

The MR app runs on a Meta Quest 3 and is implemented in Unity 6 using Meta XR SDK v85.0. The headset provides passthrough RGB images for fiducial detection and per-eye environment-depth textures with corresponding view-projection matrices for terrain reconstruction.

4.2 Geospatial Registration

During registration, the headset connects to the phone's WebSocket server over the phone's mobile hotspot. The subsequent tag detection, observation pairing, and aggregation stages are shown in Fig. 2 (top).

Tag and camera configuration We adapted Keijiro's AprilTag package for Unity[3] to process images from the Meta Passthrough Camera API[4]. Camera access uses Mixed Reality Utility Kit's `PassthroughCameraAccess`, configured for the left RGB camera at a requested resolution of 1280×960 pixels and a maximum stream rate of 60 Hz. Detection is skipped when the estimated headset linear speed exceeds 0.1 m/s or its angular speed exceeds $10\,^\circ$/s. For each image, `Timestamp` provides the camera-image timestamp, and `GetCameraPose()` returns the corresponding RGB-camera pose in Unity world coordinates.

The AprilTag detector estimates ${}^{C}\mathbf{T}_{T}$ from the detected corners, the known tag size measured between detection corners, and the camera intrinsics returned by `Intrinsics`. Because `FocalLength` and `PrincipalPoint` are expressed relative to `SensorResolution`, we convert them to `CurrentResolution` by applying the corresponding centered crop and scale.

We define the phone reference frame P to coincide with the detector-returned tag frame T: its origin is at the center of the displayed tag, $+X_P$ points right on the display, $+Y_P$ points toward the top of the phone, and $+Z_P$ points through the display, away from the viewer. We associate the Core Location position estimate with the origin of P and its reported true heading with $+Y_P$.

[3] https://github.com/keijiro/jp.keijiro.apriltag

[4] https://developers.meta.com/horizon/documentation/unity/unity-pca-documentation/

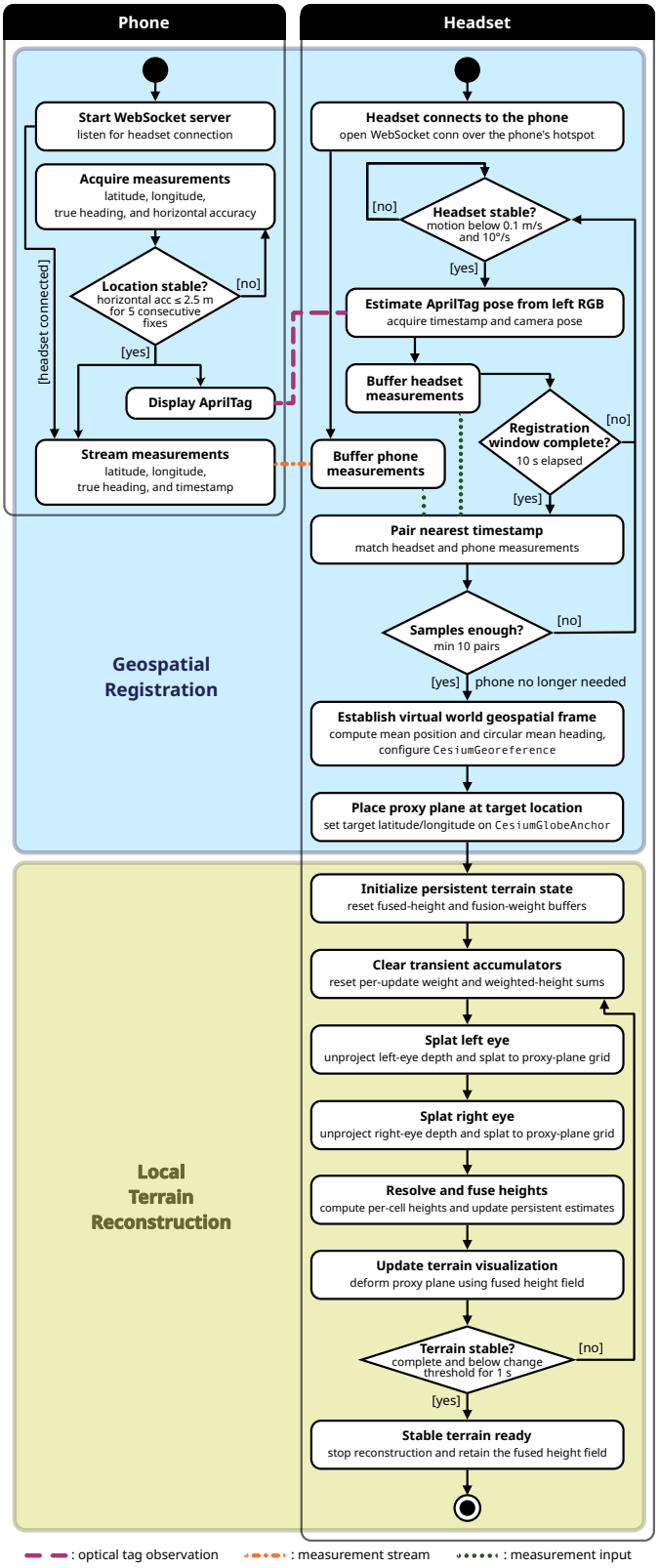


Figure 2: TeAPOT workflow: smartphone-assisted geospatial registration (top), followed by on-headset local terrain reconstruction (bottom). Lanes separate the steps executed on the phone (left) and the headset (right), and solid arrows show control flow.

Because P and T coincide, ${}^{P}\mathbf{T}_{T} = \mathbf{I}$, and Eq. (1) reduces to

$${}^{V}\mathbf{T}_{P} = {}^{V}\mathbf{T}_{C}\,{}^{C}\mathbf{T}_{T} = {}^{V}\mathbf{T}_{H}\,{}^{H}\mathbf{T}_{C}\,{}^{C}\mathbf{T}_{T}.$$

Here, ${}^{V}\mathbf{T}_{C}$ is the RGB-camera pose returned by `GetCameraPose()` and combines headset tracking ${}^{V}\mathbf{T}_{H}$ with camera-to-headset extrinsics ${}^{H}\mathbf{T}_{C}$. The complete transform chain and omitted calibration terms are shown in the supplemental material.

We calibrate neither the device-specific lever arm, a fixed position offset between the GNSS reference point and the origin of P, nor the fixed yaw misalignment between the reported true-heading reference and $+Y_P$. The former produces an approximately constant translational offset; the latter produces a lateral error that grows with target distance. Both contribute to the end-to-end error and are discussed in Sec. 6.2.

Observation pairing and aggregation Phone measurements and headset measurements are buffered separately. Phone measurements comprise position, true heading, and a transmission timestamp, whereas headset measurements comprise the detected tag pose, RGB-camera pose, and camera-image timestamp. Each headset measurement is paired with the phone measurement having the nearest timestamp. We do not estimate the clock offset between devices or separately compensate for phone-sensor acquisition, buffering, and transmission latency. Each matched pair yields one estimate of the virtual-world origin's geodetic position and true heading. We collect paired estimates for 10 s and require at least 10 valid pairs. Otherwise, collection restarts. The final origin position is obtained from the spherical mean of the geodetic estimates, and the final heading from their circular mean, which avoids wraparound at 0°/360°.

Georeferencing and proxy placement We use Cesium for Unity[5] for WGS84 georeferencing. Once the aggregated latitude, longitude, and origin heading are established, we configure a `CesiumGeoreference` with the origin coordinates. It defines the corresponding local Unity frame and provides transforms between geodetic, ECEF, and Unity coordinates. We assign zero ellipsoidal height to the origin and target points because the effect on horizontal placement is negligible over these short baselines. We then rotate the georeference about its local up axis by the estimated origin heading to align the local frame with the real-world orientation. Finally, we attach a `CesiumGlobeAnchor` to the proxy plane and set its target latitude and longitude, and ellipsoidal height to zero, placing it at the corresponding Unity position relative to the configured origin.

4.3 Local Terrain Reconstruction

We implement local terrain reconstruction as a GPU-accelerated depth-to-height-field pipeline that fuses per-eye environment-depth maps into a 2.5D terrain representation over a 1×1 m patch. As shown in Fig. 2 (bottom), each terrain update consists of four compute stages: `ClearSum`, `Splat` for the left eye, `Splat` for the right eye, and `ResolveFuse`. To bound per-frame cost on the headset, these stages are distributed across rendered frames, with at most one stage dispatched per frame. The scheduler is initialized with a target terrain-update rate of 1–4 Hz, after which the dispatch interval is adjusted online based on the measured frame time.

The splat stage consumes the per-eye environment-depth textures, `_EnvironmentDepthTexture`, and the corresponding depth-camera view-projection matrices, `_EnvironmentDepthReprojectionMatrices`, both provided by the Meta XR SDK. Each depth pixel is unprojected to world space, mapped into the proxy-plane frame, and EWA-splatted into the local grid. To support efficient concurrent accumulation on the mobile GPU, the splat weights and weighted heights are scaled to fixed-point integers and stored in integer textures. The transient accumulators are `SplatWeightSum` (`R32_UInt`) and `SplatWeightHeightSum` (`R32_SInt`). The persistent fused state is stored in `FusedWeight` and `FusedHeight` (both `R32_SFloat`), while `PrevFusedHeight` (`R32_SFloat`) stores the previous fused height field for stability checking. Since the depth textures are provided at 320×320 resolution, we use the same resolution for the local reconstruction grid. This defines the maximum available grid detail when the 1×1 m patch fills the headset's view. At the beginning of each terrain update, `ClearSum` resets the transient accumulation buffers so that the subsequent per-eye splats contribute only to that update.

After splatting both eyes, `ResolveFuse` uses the transient sums to compute the current per-cell height and update the persistent fused state. Cells with no new observation retain their previous estimate. For convergence detection, the current `FusedHeight` field is compared with the previous resolved field, `PrevFusedHeight`. This comparison produces the number of known cells and the maximum per-cell height change. These values are read back to the CPU, where reconstruction completeness is computed as the fraction of known grid cells. Completeness and maximum change drive the stability timer and automatic export logic. A reconstruction is considered stable only when completeness meets the required threshold and the maximum inter-resolve height change remains below a tolerance for a fixed hold period. In the current implementation, these thresholds are set to 100 % completeness, a maximum change of 1 cm, and a hold time of 1 s.

For visualization, we generate the proxy-plane mesh once at the same 320×320 resolution and bind `FusedHeight` to its material. The material samples the fused height field and offsets the mesh vertices along the proxy-plane normal, producing a terrain-conforming surface. For export, we read back `FusedHeight` and write the reconstructed grid as a Polygon File Format (PLY) point cloud.

5 Evaluation

We evaluate the two main technical components of TeAPOT separately: geospatial registration by absolute placement error, and local terrain reconstruction by how closely the reconstructed surface matches the terrain beneath it and by runtime performance. The reconstruction evaluation measures local surface-shape accuracy, not whether the surface was placed at the intended geospatial location.

5.1 Geospatial Registration Accuracy

5.1.1 Setup and Metrics

As ground truth for the geospatial registration evaluation, we used a dual-band L1/L5 GNSS receiver (Waveshare LC29H(DA) GPS/RTK HAT[6]) configured as an RTK rover with real-time corrections from SAPOS Saxony[7]. The nine outdoor test locations, denoted L1–L9, cover three representative outdoor MR scenarios: open sky (L2, L7, L8), building-adjacent (L3, L4, L5), and tree canopy (L1, L6, L9). A satellite view of the immediate surroundings of each site is provided in the supplemental material.

We conduct three related analyses. First, we measure static phone GNSS error at all nine locations. Second, we evaluate the complete phone-headset pipeline at three locations, one from each scenario. Third, we characterize the phone's true-heading error separately at the same three locations to assess whether its error magnitude is consistent with the observed distance-dependent behavior.

For the phone GNSS analysis, the phone recorded latitude and longitude at 15 Hz during two 10 min sessions at each site, producing 18,001–18,002 fixes per site and 162,011 fixes overall. The horizontal phone error relative to RTK is denoted ε_{hor}.

For the end-to-end analysis, we use three locations from the phone study: L2 (open sky), L3 (building-adjacent), and L6 (tree canopy). At each location, we marked five RTK-measured standing positions at 0, 1, 2, 5, and 10 m from a fixed reference point. For each trial, the user stood at one of these positions while TeAPOT estimated the user's geodetic position from the phone GNSS, phone heading, and phone-headset alignment. Each trial began with a fresh application start and consisted of one geospatial pose calculation followed by one projection of the reference point.

The horizontal error of the estimated standing position relative to its RTK ground truth is denoted ε_{pos}. Using the resulting geospatial frame, the system then projects the fixed reference point at the 0 m position. The horizontal error of the projected target relative to the RTK reference coordinate is denoted ε_{end}. Both ε_{pos} and ε_{end} are *radial* errors: each is the magnitude of a horizontal displacement from ground

[5] https://cesium.com/learn/unity

[6] https://www.waveshare.com/wiki/LC29H(XX)_GPS/RTK_HAT

[7] https://www.landesvermessung.sachsen.de/sapos/

Table 1: Phone GNSS horizontal position error (ε_{hor}).

Location	Scenario	N	P_{50}	P_{68}	P_{95}	P_{25}–P_{75}
L2	Open	18k	0.86	1.00	1.60	0.53–1.10
L3	Building	18k	2.54	3.30	4.81	1.79–3.52
L6	Tree	18k	1.64	2.08	3.88	1.21–2.30
L2, L7, L8	Open	54k	1.00	1.25	1.67	0.77–1.46
L3, L4, L5	Building	54k	2.07	2.49	4.46	1.38–2.70
L1, L6, L9	Tree	54k	1.44	1.82	3.00	0.95–1.92
All		162k	1.42	1.81	3.64	0.93–2.00

L2, L3, and L6 were also used for the heading and end-to-end evaluations.
N = sample size (rounded) from two sessions at different times (10 min each, 15 Hz). Percentile values are in meters.
Building: Building-adjacent; Open: Open sky; Tree: Tree canopy.
Per-location percentiles for all sites are provided in the supplemental material.

Table 2: End-to-end headset horizontal position error (ε_{end}).

Location	Dist	Runs	P_{50}	P_{68}	P_{95}	P_{25}–P_{75}
All	0	30	1.73	2.65	6.16	1.10–2.93
All	1	30	2.14	2.91	6.09	1.42–2.98
All	2	30	2.21	2.89	4.98	1.30–3.17
All	5	30	2.50	3.49	5.58	1.28–3.99
All	10	30	2.02	2.19	3.96	1.16–2.44
L2 (Open)	All	50	1.18	1.54	2.09	0.92–1.66
L3 (Building)	All	50	2.78	4.06	7.21	1.71–4.28
L6 (Tree)	All	50	2.73	3.30	4.89	2.04–3.61
All	All	150	2.04	2.87	5.52	1.16–3.17

Distance and percentile values are in meters.
Building: Building-adjacent; Open: Open sky; Tree: Tree canopy.
Location-by-distance percentiles are provided in the supplemental material.

truth, with the direction discarded. We therefore summarize the change with their signed difference, $\Delta\varepsilon = \varepsilon_{end} - \varepsilon_{pos}$. Thus, $\Delta\varepsilon$ describes how much the radial error changes after registration and projection, and it is not itself a vector displacement attributable to a single component.

Each location-distance cell was repeated 10 times using a fixed sequence of nominal heading targets (0°, 90°, 180°, 270°, 45°, 135°, 225°, 315°, 90°, and 270°), producing 150 trials in total. Because the headings were hand-aligned rather than mechanically enforced, the heading IDs were treated as nominal within-cell repetitions rather than exactly matched repeated-measures blocks across distances.

We test three hypotheses relevant to practical outdoor MR use:

- **H1**: The end-to-end pipeline produces a systematic signed change relative to the standing-position error, i.e., $\Delta\varepsilon \neq 0$.
- **H2**: The absolute radial-error difference ($|\Delta\varepsilon|$) increases with standing distance.
- **H3**: End-to-end error remains strongly associated with the phone's standing-position error (ε_{pos}), consistent with phone GNSS uncertainty being the primary contributor to the total error.

We therefore use a two-sided Wilcoxon signed-rank test for H1, Page's trend test with an exact Friedman robustness check for H2, and Spearman correlation for H3. For the distance tests, the 10 hand-aligned trials in each location-distance cell are aggregated to a median, producing 15 location-distance sets (3 locations × 5 distances).

5.1.2 Phone GNSS vs. RTK

Because TeAPOT inherits the phone's GNSS error, we first quantify this baseline. Table 1 reports the horizontal-error percentiles for the three end-to-end locations, the three scenario aggregates, and the full dataset. The corresponding empirical cumulative distribution function (CDF) is provided in the supplemental material. Overall, the phone achieves a median horizontal error (ε_{hor}) of 1.42 m ($P_{68} = 1.81$ m, $P_{95} = 3.64$ m).

Scenario differences were substantial. Open sky performed best ($Mdn = 1.00$ m, $P_{95} = 1.67$ m), building-adjacent performed worst ($Mdn = 2.07$ m, $P_{95} = 4.46$ m), and tree canopy lay in between ($Mdn = 1.44$ m, $P_{95} = 3.00$ m). The median phone errors at the three end-to-end test locations were 0.86 m at L2, 2.54 m at L3, and 1.64 m at L6. These descriptive results provide the environmental context for interpreting the end-to-end evaluation.

5.1.3 End-to-End Positioning Accuracy

Across all 150 trials, the full end-to-end error (ε_{end}) has a median of 2.04 m ($P_{68} = 2.87$ m, $P_{95} = 5.52$ m) (see Tab. 2). Performance again follows the environmental quality of the phone baseline: the open-sky location L2 gives a median end-to-end error of 1.18 m, while the building-adjacent and tree-canopy locations L3 and L6 give medians of 2.78 m and 2.73 m, respectively.

The change in horizontal error after registration and projection is much smaller than the total end-to-end error. Supporting H1, the signed difference $\Delta\varepsilon$ is positive overall but small in magnitude ($Mdn = +0.042$ m, $IQR = [-0.014, +0.158]$ m; Wilcoxon signed-rank $p < .001$), as also shown in Fig. 3C. The same conclusion holds after aggregating to 15 location-distance medians ($p < .001$), indicating that the effect is not driven by a few outlier trials. Importantly, $\Delta\varepsilon$ is already nonzero at 0 m ($Mdn = +0.019$ m), where no projection distance is involved. This suggests that part of the signed change comes from manual user positioning at the RTK-measured standing position and from the end-to-end registration process, rather than from heading projection alone.

H2 is supported more clearly when the radial-error difference is evaluated by absolute magnitude rather than sign. The signed median $\Delta\varepsilon$ increases from +0.019 m at 0 m to +0.183 m at 10 m, as summarized in Fig. 3C, but the trend is stronger in $|\Delta\varepsilon|$: its median grows from 0.028 m at 0 m and 0.036 m at 1 m to 0.254 m at 5 m and 0.475 m at 10 m. This stronger distance effect is visible both in the distance-stratified CDFs in Fig. 3B and the per-distance summary in Fig. 3D. Per-distance counts, full interquartile ranges, and P_{95} values are reported in the supplemental material. Page's trend test on the 15 location-distance medians confirms a significant monotone increase with distance ($L = 163$, exact $p < .001$), supported by the exact Friedman test ($\chi^2(4) = 10.933$, $p = .003$, $W = .91$).

H3 is also supported. Figure 3A shows that ε_{end} closely tracks ε_{pos} across locations and standing distances. The overall Spearman correlation is $\rho = 0.932$ ($p < .001$), indicating that the phone's GNSS uncertainty is likely the primary component of the end-to-end error in this study, although part of the association also reflects environmental differences between locations. Consistent with this, the median standing-position error (ε_{pos}) and end-to-end error (ε_{end}) remain close at each location: 1.13 m vs. 1.18 m at L2, 2.69 m vs. 2.78 m at L3, and 2.63 m vs. 2.73 m at L6. Full ε_{pos} percentiles per location and standing distance are reported in the supplemental material.

5.1.4 Phone Heading Accuracy

A heading error rotates the projected target, producing displacement that increases with range and may contribute to the increase in $|\Delta\varepsilon|$ observed for H2. Because the end-to-end trials did not record per-trial reference headings and $\Delta\varepsilon$ compares radial magnitudes rather than heading-induced displacement vectors, we characterized the same phone's true heading in a separate experiment. We use the resulting heading-error distribution only to assess consistency with the distance trend, and it does not isolate heading from other registration errors.

We characterized heading accuracy at the three end-to-end evaluation sites: L2, L3, and L6. At each site, we collected reciprocal A-to-B and B-to-A blocks on the same day. For each block, we re-surveyed both endpoints with RTK to define the reference bearing. A and B therefore denote nominal endpoints, because RTK uncertainty and repositioning caused coordinate differences between reciprocal blocks. The points were separated by 25.75–31.27 m to reduce the effect of position uncertainty on the reference bearing. The A-to-B and B-to-A bearings were 245.82° and 65.86° at L2, 294.68° and 111.92° at L3, and 195.71° and 16.03° at L6.

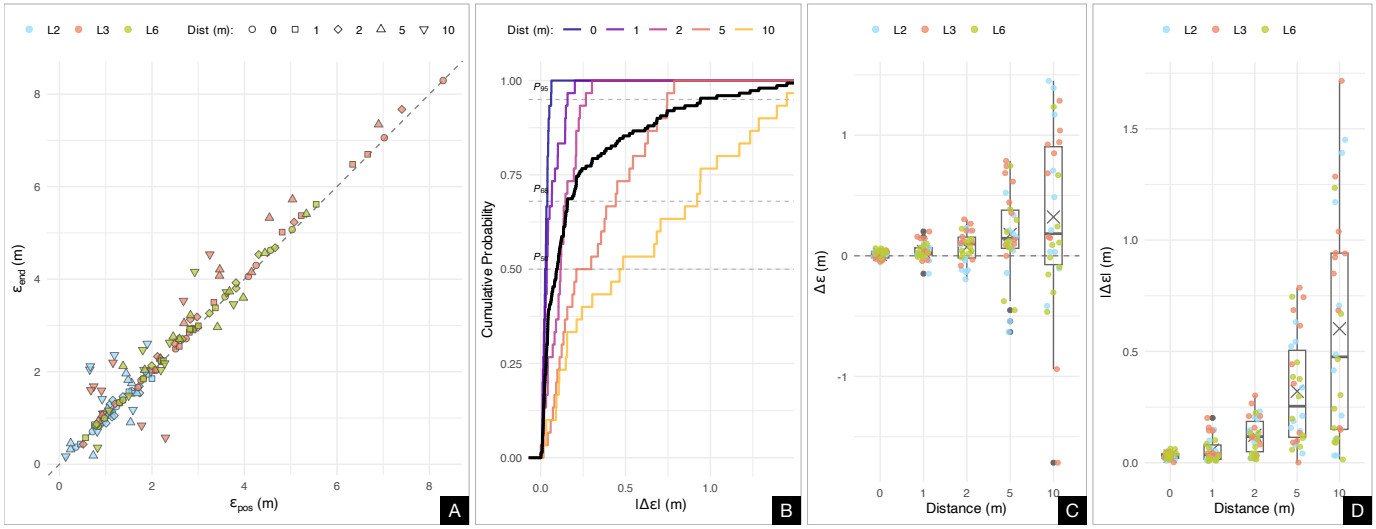


Figure 3: End-to-end error behavior across the 150 trials. In A, C, and D, color encodes the location (L2, L3, L6). (A) ε_{end} plotted against ε_{pos}, with marker shape encoding standing distance. The dashed diagonal indicates $\varepsilon_{\text{end}} = \varepsilon_{\text{pos}}$, i.e., no change in horizontal error after registration and projection. (B) CDF of $|\Delta\varepsilon|$ per standing distance, with the black curve pooling all 150 trials and the dashed horizontal lines marking the percentile levels. The separation of the curves at standing distances beyond 1 m mirrors the monotone distance effect tested for H2. (C) Signed $\Delta\varepsilon$ per standing distance, shown as boxplots over jittered per-trial points, with $\times$ marking the mean and the dashed line marking zero. It remains small overall but grows at larger distances. (D) $|\Delta\varepsilon|$ increases with distance, matching the direction tested for H2 with Page's test and checked with the Friedman test.

Table 3: Session-level phone true-heading accuracy.

Loc	Direction	Sessions	Bias	MAE	RMSE	P_{50}	P_{68}	P_{95}
L2	A to B	10	−3.52	3.52	3.61	3.48	3.75	4.62
L2	B to A	10	3.94	3.94	4.05	3.95	4.18	5.29
L3	A to B	10	−5.22	5.22	5.30	5.46	5.94	6.12
L3	B to A	10	3.31	3.31	3.53	3.40	4.13	4.90
L6	A to B	10	−2.94	2.94	3.08	3.11	3.62	3.99
L6	B to A	10	7.56	7.56	7.70	7.80	8.20	9.73
All		60		4.42	4.81	4.08	5.06	8.19

Each session contributes the median reported heading over its one-minute analysis window. All angular values are in degrees. Bias is the signed circular mean. Mean absolute error (MAE), root mean square error (RMSE), and percentiles are computed from the absolute session-level heading error ε_{hdg}.

The phone was placed horizontally with its screen facing upward. Using an aluminum rail and a right-angle set square, we aligned the phone's longitudinal axis with a laser rangefinder beam and oriented the phone so that the direction from its center toward its top edge matched the beam direction toward the opposite RTK-surveyed endpoint. This aligned the Core Location heading reference used by the system with the surveyed baseline. Before each session, we moved the phone through a figure-eight motion to allow the compass calibration to update, then repositioned it and repeated the alignment. We logged the reported true heading ψ_P during 10 stationary sessions per site-direction combination, each recording roughly two minutes of data. From each session, we analyzed the one-minute window beginning when the phone first reported a stable location fix. Across the 60 analysis windows, the log contained 54,020 rows, but each window contained only 1–5 distinct heading values. We therefore retained the median over each window, yielding 60 session-level estimates across six site-direction combinations. Heading error was defined as the signed angular difference between the session median and the RTK bearing, wrapped to $[-180^\circ, 180^\circ)$. Its magnitude, the absolute session-level heading error, is denoted ε_{hdg}.

Table 3 reports the aggregate and per-site-direction heading-error statistics. The CDF of the 60 session-level absolute errors is provided in the supplemental material. Across all 60 session-level estimates, the median absolute heading error (ε_{hdg}) is 4.08° ($P_{68} = 5.06^\circ$, $P_{95} = 8.19^\circ$, RMSE $= 4.81^\circ$). Within-session variation is small, with the largest standard deviation being 0.81°. The signed circular biases for A-to-B and B-to-A are -3.52° and $+3.94^\circ$ at L2, -5.22° and $+3.31^\circ$ at L3, and -2.94° and $+7.56^\circ$ at L6. Because measurement direction, collection order, and reference bearing changed together at each site, these offsets are descriptive and do not isolate a direction effect. Additional per-session diagnostics are provided in the supplemental material.

A heading error θ produces a target displacement at range d of

$$h(d, \theta) = 2d \sin(|\theta|/2) \approx d|\theta|,$$

with θ expressed in radians. However, $|\Delta\varepsilon|$ is the absolute difference between two radial error magnitudes, not the heading-induced displacement error. The reverse triangle inequality gives $|\Delta\varepsilon| \leq \|\mathbf{e}_{\text{end}} - \mathbf{e}_{\text{pos}}\|$, but $h(d, \theta)$ accounts for only the heading-induced component of $\mathbf{e}_{\text{end}} - \mathbf{e}_{\text{pos}}$, which also absorbs standing-position, transform-chain, and tag-pose errors. It is therefore neither an upper nor a lower bound on $|\Delta\varepsilon|$, and we use it only as a reference scale. The corresponding plot is provided in the supplemental material.

For reference, at every nonzero projection distance, the observed median $|\Delta\varepsilon|$ was below the median heading-induced displacement computed from the independently measured heading errors. At 10 m, the median heading-induced displacement is 0.71 m, compared with an observed median $|\Delta\varepsilon|$ of 0.475 m. At the same projection distance, the observed $|\Delta\varepsilon|$ has $P_{95} = 1.42$ m and root mean square (RMS) $= 0.78$ m, while the heading-induced displacement has $P_{95} = 1.43$ m and RMS $= 0.84$ m. The measured heading errors could therefore contribute to the distance trend. This comparison does not identify heading as the primary cause because the radial metric discards error direction, the two evaluations were conducted separately, and other registration errors were not controlled.

5.2 Local Terrain Reconstruction Accuracy and Runtime Performance

5.2.1 Setup and Metrics

We next evaluate local terrain-reconstruction accuracy and runtime performance. For each trial, the headset reconstructs a 1×1 m height field of the ground surface at a specified location. To ensure repeatability, we place a printed AprilTag at the center of the patch and use it to trigger reconstruction. This helps align repeated runs to the same physical 1×1 m patch. After the terrain reaches full completeness and remains stable for 1 s, the headset exports it as a point cloud. The fused height

Table 4: Post-registered local terrain reconstruction accuracy and runtime performance across terrain types.

	Post-registered local surface error					Runtime and stability					
Terrain	**Median** (cm)	P_{95} (cm)	**RMSE** (cm)	**Mean** (cm)	**Valid comparison area** (%)	**TTS** (s)	**Resolve** (Hz)	**GPU** (ms)	**FPS**	**Stale frames**	**Max consec. stale**
Bare soil	0.6 [0.5, 0.7]	1.0 [0.9, 1.1]	2.1 [1.6, 2.3]	0.2 [0.2, 0.2]	96.00 [96.00, 96.00]	1.1 [1.1, 1.2]	4.6 [4.5, 4.6]	5.5 [5.2, 5.7]	88.5 [86.1, 89.4]	4.5 [1.6, 6.8]	3.5 [2.0, 5.8]
Mixed	0.8 [0.7, 0.8]	1.2 [1.2, 1.3]	2.4 [2.3, 2.5]	0.0 [−0.1, 0.0]	98.10 [98.10, 98.18]	1.3 [1.1, 1.6]	4.4 [4.3, 4.6]	5.7 [5.3, 5.9]	88.5 [88.5, 88.9]	3.0 [2.6, 5.4]	3.0 [2.3, 4.0]
Grass	0.5 [0.4, 0.6]	1.2 [1.0, 1.3]	2.5 [2.1, 3.3]	−0.2 [−0.6, −0.1]	93.60 [93.53, 93.60]	1.1 [1.1, 1.3]	4.5 [4.2, 4.6]	5.5 [4.8, 5.7]	89.0 [88.1, 89.4]	8.8 [4.5, 12.8]	5.5 [2.3, 19.8]
All	0.6 [0.5, 0.8]	1.1 [1.0, 1.3]	2.3 [2.1, 2.6]	0.0 [−0.2, 0.2]	96.00 [93.60, 98.10]	1.1 [1.1, 1.4]	4.5 [4.3, 4.6]	5.6 [5.2, 5.8]	88.5 [88.0, 89.4]	5.3 [3.0, 11.5]	3.5 [2.0, 6.8]

Reconstruction completeness was 100 % for all trials. Values are reported as median [Q1, Q3]. Results are based on 10 trials per terrain type (30 trials in total). Accuracy metrics are 2.5D height differences after rough alignment and Iterative Closest Point (ICP): Median, P_{95}, and RMSE are reported as absolute height differences, while Mean is reported as the mean signed height difference. These metrics do not include absolute geospatial placement error.

field is also visualized in situ by deforming the proxy-plane mesh and rendering it with a simple shader.

The same surface is also captured using a LiDAR-equipped phone with Scaniverse[8], which we treat as ground truth. We compare the reconstructed terrain against this reference in CloudCompare[9]. First, the reference mesh is sampled into a dense point cloud of up to one million points. The reconstructed and reference point clouds are then brought into a common coordinate frame by rough alignment followed by ICP. Because both surfaces can be treated as 2.5D height fields, we use CloudCompare's Compute 2.5D Volume tool with a grid step of 0.3125 cm, corresponding to the 1×1 m patch discretized into 320×320 cells. This yields the valid comparison area, mean signed error, RMSE, and an absolute height-difference map, from which we report the median and the 95th percentile.

For each reconstruction, we log time to stability (TTS) and the resolve rate. We define TTS as the elapsed unscaled time from reconstruction start to automatic export, including the required 1 s stable hold period. Stability is evaluated after each resolve using two conditions: all terrain cells must be known, and the maximum per-cell height change relative to the previous resolved terrain must not exceed 1 cm. The average resolve rate is computed as the number of executed resolve steps divided by TTS, yielding the effective realized update frequency.

Additionally, we run the OVR Metrics Tool[10] during reconstruction to obtain other runtime metrics, including GPU time, frames per second (FPS), stale frames, and maximum consecutive stale frames. As defined by the tool, the stale-frame metrics reflect display-level frame-delivery quality. A stale frame occurs when a new frame is not delivered on time and the previous frame is reused, whereas the maximum consecutive stale-frame metric denotes the longest uninterrupted streak of such events. We include these metrics because they help indicate performance issues that may degrade user comfort or cause motion sickness.

We evaluate three locations representing common terrain conditions: bare soil, mixed terrain, and grass (see Fig. 1C–1E, respectively). At each location, we repeated the reconstruction 10 times, resulting in 30 trials. In each trial, the user stood approximately 2 m from the tag. Each trial began with a fresh application start, followed by one reconstruction run and terrain export. Across the 10 trials, the user position was varied around the target patch to cover different viewing angles, with reconstruction initiated by looking at the tag.

5.2.2 Accuracy

Table 4 summarizes local terrain accuracy over 30 trials. The reconstructed terrain achieves a median absolute error of 0.6 cm [0.5, 0.8], a P_{95} absolute error of 1.1 cm [1.0, 1.3], and an RMSE of 2.3 cm [2.1, 2.6], with a valid comparison area of 96.00 % [93.60, 98.10]. The mean signed error is close to zero at 0.0 cm $[-0.2, 0.2]$, indicating little systematic vertical bias. The valid comparison area indicates the fraction of cells for which both the reconstructed and reference surfaces provide comparable height values. Differences across terrain types are modest, with grass showing the lowest median absolute error, bare soil the lowest P_{95} absolute error and RMSE, and mixed terrain the highest valid comparison area. These results indicate centimeter-level post-registered local terrain reconstruction accuracy across all tested terrain types. They do not measure absolute geospatial placement of the reconstructed patch. For qualitative inspection, we render the reconstructed terrain in CloudCompare using EDL shading. Panels C–E of Fig. 1 show the real-terrain mesh, reconstructed terrain, and resulting absolute height-difference map for bare soil, mixed terrain, and grass, respectively. The height-difference color scale, ranging from 0.0–6.0 cm, is shown in Fig. 1F.

5.2.3 Runtime Performance

Table 4 also reports runtime performance over 30 trials. Reconstruction reaches stability after 1.1 s [1.1, 1.4] and achieves a resolve rate of 4.5 Hz [4.3, 4.6]. A resolve rate of 4.5 Hz corresponds to one fused terrain update approximately every 0.22 s, allowing the terrain estimate to refresh several times per second. GPU time is 5.6 ms [5.2, 5.8], while FPS remains at 88.5 [88.0, 89.4], suggesting that reconstruction remains lightweight relative to the application frame budget. The method records 5.3 [3.0, 11.5] stale frames, with a median maximum consecutive stale-frame count of 3.5 [2.0, 6.8]. While stale frames should ideally remain at zero, the observed stale-frame counts and short maximum consecutive streaks suggest that running terrain reconstruction introduces occasional frame-delivery disruptions, but not prolonged visible stalls. Runtime performance is similar across terrain types, with grass showing the highest stale-frame counts, mixed terrain the lowest, and bare soil the highest resolve rate. These results reflect terrain reconstruction performance under our prototype implementation, including the lightweight in situ terrain visualization shown in Fig. 1B, but not the cost of more advanced terrain rendering, which depends on the chosen mesh representation and shader.

6 Discussion

6.1 Design Implications for Outdoor MR

Across all headset trials, TeAPOT's end-to-end error (ε_{end}) is strongly correlated with the phone standing-position error (ε_{pos}). At the three headset locations, their medians differ by only 5–10 cm. Thus, registration and projection change the horizontal error much less than the meter-scale error already present in the standing-position estimate.

Registration accuracy nevertheless depends strongly on the environment. In open-sky conditions, the phone baseline has a median horizontal error of approximately 1 m, and the headset end-to-end median at L2 is 1.18 m. The building-adjacent and tree-canopy sites instead produce median end-to-end errors of approximately 2.7–2.8 m. TeAPOT is therefore best suited to meter-tolerant outdoor MR, especially in open-sky conditions, such as exploratory visualization of smoothly varying environmental fields, ground-attached annotations, and area-scale overlays. It is not a replacement for survey-grade registration in tasks requiring submeter or centimeter alignment.

The distance-dependent behavior of $|\Delta\varepsilon|$ suggests a clear usage guideline. Its median is 0.028 m at 0 m and 0.036 m at 1 m, but increases to 0.254 m at 5 m and 0.475 m at 10 m. Users should therefore register near the content being inspected and re-register when standing far from it. Minimizing the projection distance limits the effect of orientation and other distance-dependent registration errors.

[8] https://dev.scaniverse.com

[9] https://www.cloudcompare.org

[10] https://developers.meta.com/horizon/documentation/unity/ts-ovrmetricstool

For terrain reconstruction, the results suggest that the method is suitable for centimeter-level local surface agreement within a 1×1 m patch. This level of agreement supports terrain-conforming rendering, although occasional local outliers remain. However, these results do not measure absolute geospatial placement or end-to-end global terrain accuracy.

The terrain estimate stabilizes after approximately 1.1 s while resolving several times per second. The low GPU time and high FPS indicate modest overhead in the current prototype. However, the nonzero stale-frame counts indicate that frame-delivery disruptions occurred during reconstruction. These disruptions should be considered when combining reconstruction with heavier rendering or larger reconstruction regions.

6.2 Limitations

The geospatial registration evaluation deliberately reflects realistic use: the phone was handheld, and headings were aligned only to nominal targets. We therefore cannot interpret the heading-trial IDs as exact repeated-measures blocks across distances. Consequently, the distance-effect tests aggregate the 10 trials in each location-distance cell to a median. This analysis does not isolate heading bias, phone-handling variation, headset drift, tag-pose error, or standing-position error. Moreover, $\Delta\varepsilon$ is a difference between radial magnitudes, not a vector decomposition of those sources.

Even at zero projection distance, the median signed $\Delta\varepsilon$ is nonzero, indicating that part of the radial-error difference arises from the end-to-end registration workflow and manual positioning at the RTK-surveyed standing point rather than heading-driven projection alone. A controlled experiment should separately quantify standing-position, transform-chain, and projection errors.

TeAPOT uses the camera intrinsics and camera-headset extrinsics provided by the Meta XR SDK, but does not calibrate the phone-specific lever arm between the GNSS reference point and the displayed tag, or any fixed yaw misalignment between the reported true-heading reference and the tag frame. The omitted lever arm is bounded by the phone dimensions (see supplemental material) and therefore introduces an approximately constant centimeter-scale translational offset. By contrast, a fixed heading-to-tag misalignment, heading bias, and angular errors in the camera pose, tag-pose estimate, or frame alignment rotate the recovered geospatial frame, causing placement error to increase with target distance. Together with the phone's meter-scale GNSS uncertainty, these uncalibrated effects limit registration accuracy and remain included in the reported end-to-end error. At the measured median absolute heading error of 4.08°, the heading-induced displacement at a projection distance of 10 m is 0.71 m. At the measured P_{95} of 8.19°, it is 1.43 m. Thus, the measured heading errors can produce displacements ranging from submeter to meter scale at this distance.

The end-to-end evaluation covers only three locations and one phone-headset combination, spanning open-sky, building-adjacent, and tree-canopy conditions. The building-adjacent site L3 had the highest median end-to-end error at 2.78 m, but the current evaluation does not isolate the relative contributions of GNSS, heading, phone-headset alignment, and headset tracking. Denser built environments, such as narrow streets, enclosed squares, and building-lined corridors, may further degrade both GNSS and compass performance through stronger multipath and magnetic disturbance from vehicles and reinforced structures. We therefore do not claim reliable operation in dense built environments without an additional localization source, such as VPS.

The separate compass characterization is likewise limited to one phone and six site-direction combinations at three sites. At each site, the A-to-B block was followed by the B-to-A block on the same day, with a separately surveyed reference bearing for each direction. Because direction, measurement order, and reference bearing changed together, the observed offsets cannot be attributed specifically to environmental magnetic interference. The compass characterization and end-to-end evaluation were also conducted in separate campaigns, making their comparison a magnitude-level consistency check rather than a paired analysis.

Terrain reconstruction was evaluated within a 1×1 m patch, and future evaluation should examine larger regions and a broader range of terrain conditions. The lightweight 2.5D height field cannot represent multiple heights at the same planar location, such as overhangs or shrubs. The prototype also does not explicitly filter transient occluders such as body parts or passing people, which may be fused temporarily until later depth observations overwrite them.

Finally, the runtime evaluation covers the lightweight prototype, with terrain reconstruction and simple in situ visualization running together. It therefore does not isolate reconstruction overhead or establish performance for applications with more complex rendering, interaction, or larger reconstruction regions.

6.3 Future Work

Future work should reduce the distance-dependent error growth observed at larger standing distances. Possible directions include device-specific lever-arm and heading-reference calibration, rigid phone mounting, per-session heading-bias estimation, fusion with additional orientation cues, and interaction techniques that guide users to register near the target.

A systematic comparison with other modern smartphones, particularly flagship Android devices with dual- or multi-frequency GNSS, would clarify TeAPOT's sensitivity to specific sensor stacks.

For local terrain reconstruction, the method could be extended to cover tiled terrain regions and fuse existing Digital Elevation Model/Digital Terrain Model data while retaining the ability to operate in unmapped areas. On LiDAR-equipped phones, phone-captured depth could serve as an additional terrain source, while headset depth remains the default to preserve compatibility with non-LiDAR phones.

7 Conclusion

TeAPOT combines smartphone-assisted geospatial registration with on-headset local terrain reconstruction using commodity hardware. Across 150 outdoor trials, the median end-to-end horizontal placement error was 2.04 m ($P_{95} = 5.52$ m), and the final target error remained strongly associated with the standing-position error. A separate 60-session characterization yielded a median absolute true-heading error of 4.08° ($P_{95} = 8.19^\circ$). The heading results are consistent with the increase in added error at larger standing distances, although the evaluation does not isolate heading from the other registration terms. After rigid alignment to the reference surfaces, the reconstructed terrain had a median absolute height error of 0.6 cm ($P_{95} = 1.1$ cm). This measures local surface-shape agreement rather than centimeter-level absolute placement. TeAPOT is therefore suited for terrain-conforming outdoor MR applications that tolerate meter-scale global placement error, particularly in open-sky conditions, rather than survey-grade registration.

Supplemental Materials

All supplemental materials are publicly available at doi.org/10.5281/zenodo.21468153, released under a CC BY 4.0 license. In particular, they include (1) the raw and processed evaluation data and logs, (2) the R analysis scripts used to create all tables and figures, (3) the supplementary videos showing the evaluation setup and diagnostic visualizations, and (4) the authors' version of this paper with all appendices. The TeAPOT source code for the headset and phone applications is separately available at github.com/PiBu-Labs/TeAPOT.

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